INSTITUTIONAL MANAGEMENT PLATFORM

# School Management System

A comprehensive web-based platform for school administration, academic tracking, student admissions, fee collections, staff management, and parent-teacher communication.

**USER ACCESS**

**5 Distinct Portals (Admin, Teacher, Parent, Student, Multi-School Trust)**

**CORE MODULES**

**14 Functional Modules for Daily School Operations**

**ARCHITECTURE**

**Multi-Tenant Cloud, High-Speed Excel Ingestion & Role-Based Security**

Document Type: **System Overview & Client Specification**
Deployment: **Cloud / Hosted SaaS**

Release Version: **2.0**
Date: **October 2026**

SYSTEM OVERVIEW

# How the Platform Works

The Volpehub School Management System is designed to centralize and automate school administration. It connects administrative staff, educators, parents, and students through dedicated web interfaces backed by a centralized, secure relational database.

## Core Operational Principles

### 1. Multi-Tenant School Isolation

Each school has its own independent database context. Student records, financial transactions, and employee profiles are completely separated to prevent data mixing between institutions.

### 2. Role-Based Access Control

Every user account is assigned an explicit role: Super Admin, School Admin, Teacher, Parent, or Student. Users only have access to information and actions relevant to their responsibilities.

### 3. High-Volume Data Ingestion

Built-in bulk Excel import allows school administrators to upload hundreds or thousands of students and staff members in single batches, automatically creating user accounts and login slips.

> **White-Label Branding:** Each school can configure its own logo, custom school colors, official contact details, and custom subdomain so the system aligns with the school's identity.

## Technical Specifications

| COMPONENT | TECHNOLOGY | FUNCTION |
|---|---|---|
| **Frontend Interface** | React, TypeScript, Redux Toolkit, Tailwind CSS | Responsive single-page application that works seamlessly on desktop, laptop, tablet, and mobile browsers. |
| **Backend API** | Python, Django REST Framework, Daphne ASGI | Processes business logic, validates data, manages authentication, and handles real-time background notifications. |
| **Database** | PostgreSQL | ACID-compliant relational database ensuring financial ledger accuracy, student history preservation, and encrypted storage. |
| **Performance Caching** | Redis Fast-Cache Layer | Caches frequent read queries such as rosters, dashboard statistics, and timetables to ensure fast response times. |
| **Document & Media Storage** | Database Binary Storage & Media Vault | Stores student admission documents, verified certificates, teacher credentials, and school logos securely without broken links. |
| **Notifications** | Firebase Cloud Messaging (FCM) | Delivers push notifications to mobile devices and browsers for attendance alerts, exam announcements, and fee notices. |

USER ACCESS

# The 5 Dedicated User Portals

The platform includes 5 distinct interfaces tailored to the daily responsibilities of each user group in an educational institution:

**1. SUPER ADMIN PORTAL**

### Educational Trust & Multi-Branch Management

For trust directors and group administrators managing multiple campuses.

- Cross-school overview dashboard with aggregated headcount and financial metrics.
- Create and provision new school branches and subdomains.
- Assign school administrator accounts and manage system credentials.
- Configure platform-wide settings and monitor school activity.

**2. SCHOOL ADMIN PORTAL**

### Institutional Administration & Operations

For Principals, Vice Principals, and Administrative Office staff.

- Real-time dashboard: Active student headcount, faculty count, fee collection progress.
- Student Information System (SIS) management, student profiles, and class allocations.
- Online admissions processing, applicant review, and digital document verification.
- Fee structure definition, receipt generation, outstanding dues, and staff payroll.
- Timetable creation, exam scheduling, report card generation, and bus route tracking.

**3. TEACHER PORTAL**

### Classroom Management & Student Evaluation

For subject teachers, class teachers, and department heads.

- Class attendance marking with daily present, absent, and leave statuses.
- Homework assignment creation with instructions, due dates, and attachments.
- Exam marks entry for assigned subjects with automatic percentage calculation.
- Daily class timetable and period schedule view.
- Direct messaging with parents and access to personal salary disbursement slips.

**4. PARENT PORTAL**

### Ward Monitoring & Academic Transparency

For mothers, fathers, and authorized guardians.

- Multi-child switcher: View and manage multiple enrolled siblings under one login.
- Daily attendance status, monthly attendance rates, and absence notifications.
- Fee statement: Check total fees, paid amounts, pending balances, and download receipts.
- Exam report cards, subject grades, teacher remarks, and GPA progress.
- Daily homework tracker, school circulars, and direct chat with teachers.

**5. STUDENT PORTAL**

### Student Academic Desk

For enrolled students to track their personal academic obligations.

- Class timetable showing daily periods, subject slots, and assigned teachers.
- Active homework assignments, project instructions, and submission deadlines.
- Exam timetables, printable examination hall tickets, and published report cards.
- Personal document vault (certificates, identity cards) and school notices.

FUNCTIONAL MODULES

# Core Administrative Modules (Part 1)

### 1. Online Admissions & Digital Applications

Manages the end-to-end student intake process digitally.

- **Shareable Application Links:** Create public admission links with custom expiry dates and grade limits.
- **Document Collection:** Parents upload Birth Certificates, Transfer Certificates, and Aadhaar cards digitally.
- **Application Review:** Administration reviews submitted profiles with an integrated document previewer.
- **One-Click Admission:** Approved applicants automatically transfer into active student records with assigned roll numbers.

### 2. Student Information System (SIS) & Bulk Excel Ingestion

Central master registry for all enrolled students.

- **Batch Excel Import:** Import existing school data from Excel spreadsheets with column mapping and validation checks.
- **Automatic Account Creation:** Automatically generates separate student and parent login accounts with temporary passwords.
- **Comprehensive Student Profile:** Stores personal details, parent contacts, medical notes, blood group, and photos.
- **Annual Class Promotion:** Promotes students to the next class while maintaining historical archives of prior terms.

### 3. Academic Setup & Class Timetables

Organizes the academic structure and daily period allocations.

- **Academic Hierarchy:** Configure Academic Cycles (e.g., 2026-27), Classes, Sections (A, B, C), and Subjects.
- **Subject Assignments:** Link core subjects and electives to specific classes and assign designated subject teachers.
- **Period Scheduler:** Define daily period slots, break timings, and classroom assignments.
- **Conflict Checking:** Prevents scheduling the same teacher across two different classes during the same period.

### 4. Daily Student Attendance Management

Tracks classroom attendance with historical audit logging.

- **Rapid Marking:** Class teachers can mark an entire class present in one click, then adjust absent or late students.
- **Real-Time Percentage:** Automatically computes cumulative attendance percentages for report cards.
- **Parent Notifications:** Automatically notifies parents when their child is recorded absent.
- **Audit Logging:** Logs the exact timestamp and teacher identity for every attendance entry submitted.

FUNCTIONAL MODULES

# Core Administrative Modules (Part 2)

### 5. Fee Billing, Collections & Staff Payroll

Manages student tuition invoicing and teacher payroll disbursements.

- **Fee Structure Configuration:** Define class-wise tuition fees, lab fees, admission charges, and transport costs.
- **Official Fee Receipts:** Generates numbered, printable fee receipts with institution branding for every transaction.
- **Outstanding Dues Tracking:** Clear dashboards showing collected revenue vs. unpaid fees by grade or student.
- **Staff Salary Vouchers:** Record teacher salary payments, deductions, and generate official payment slips.

### 6. Examinations, Hall Tickets & Report Cards

Coordinates term assessments and student academic evaluation.

- **Exam Datesheet Publisher:** Publish exam schedules with subject dates, time slots, and maximum marks.
- **Printable Hall Tickets:** Automatically generate exam admit cards complete with candidate details and photos.
- **Controlled Marks Entry:** Teachers submit subject marks with administrator lock/unlock deadlines to avoid tampering.
- **Report Card Generation:** Automatically calculates totals, letter grades, percentages, and performance remarks.

### 7. Transport & Bus Route Management

Oversees school vehicles, routes, and passenger rosters.

- **Route & Stop Setup:** Define bus routes, pickup stops, and morning/afternoon scheduled times.
- **Vehicle & Driver Directory:** Store driver names, contact numbers, vehicle registration, and conductor details.
- **Student Bus Allocation:** Assign students to specific routes and stops for accurate passenger manifests.
- **Parent Route Visibility:** Parents can view their child's assigned route, driver contact, and scheduled timings.

### 8. Communications, Circulars & School Calendar

Facilitates school-wide announcements and direct parent-teacher messaging.

- **Broadcast Notices:** Distribute school-wide circulars, holiday notices, and emergency announcements.
- **Direct In-App Chat:** Secure parent-teacher communication channel without sharing personal mobile numbers.
- **Event Calendar:** Publish institutional events, sports meets, parent-teacher meetings, and examination dates.
- **Push Notifications:** Delivers instant alerts directly to smartphones and web browsers.

IMPLEMENTATION

# Deployment, Onboarding & Security

The system is engineered for smooth onboarding with minimal administrative disruption. Existing school data can be imported directly using standardized Excel spreadsheets.

## 4-Stage Onboarding Process

| STAGE | ACTION ITEMS | DELIVERABLES |
|---|---|---|
| **Stage 1** | **System Setup & Branding** | Provision school tenant, upload school logo, configure brand colors, and initialize the active academic year. |
| **Stage 2** | **Master Data Ingestion** | Import classes, sections, student roster, and staff directory via Excel templates. Credentials generated automatically. |
| **Stage 3** | **Operational Configuration** | Configure class timetables, assign subject teachers, and set up quarterly fee structures with billing amounts. |
| **Stage 4** | **Staff Briefing & Go-Live** | Distribute login slips to teachers and parents, activate daily attendance, and commence live portal operations. |

## Data Security & Access Controls

### Data Privacy & Protection

- **Isolated Tenant Data:** Complete database separation prevents any institution from accessing another school's records.
- **Credential Vault:** User passwords are encrypted using secure cryptographic hashing algorithms.
- **Session Token Rotation:** Short-lived access tokens with HttpOnly refresh cookies prevent session hijacking.

### Administrative Control

- **Temporary Password Enforcement:** Requires new users to set a private password upon first login.
- **Audit Logging:** Administrative actions such as fee collections and attendance edits are permanently logged.
- **Automatic Backups:** Central PostgreSQL database supports automated snapshot and recovery procedures.

## Comparison: Traditional Records vs. Volpehub School ERP

| OPERATIONAL AREA | TRADITIONAL MANUAL METHODS | VOLPEHUB SCHOOL ERP |
|---|---|---|
| **Student Records** | Physical register books and scattered desktop files. | Centralized searchable database with instant profile retrieval. |
| **Fee Invoicing** | Handwritten paper receipt books prone to calculation errors. | Automated billing, numbered PDF receipts, and live dues tracking. |
| **Parent Updates** | Paper diary notes and periodic parent meetings. | Instant real-time attendance, homework, and fee alerts online. |
| **Report Cards** | Teachers manually calculate and write scores by hand. | Automated percentage, grade, and GPA calculation with PDF export. |

**Demonstration & Trial:** Demonstrations can be scheduled directly with pre-populated school branches to experience the live admin console, teacher portal, and parent interface.